

JEE MAIN REVISION MASTER: TRIGONOMETRIC RATIO AND IDENTITIES

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 69 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Trigonometric Ratio And Identities — Core Principles & Governing Laws	37 PYQs	Core Principle
Very High	Trigonometric Ratio And Identities — Standard Calculations & Problem Archetypes	32 PYQs	Quantitative Analysis
Standard	Trigonometric Ratio And Identities — Applied Cases, Reactions & Traps	0 PYQs	Applied Topics

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Trigonometric Ratio And Identities — Core Principles & Governing Laws	$\sin\theta + \cos\theta = \frac{1}{2}$, $\cos(x + y) = \cos x \cos y - \sin x \sin y$, $\sin^2 x + \cos^2 x = 1$
Trigonometric Ratio And Identities — Standard Calculations & Problem Archetypes	$\frac{2\sin x}{1 + \cos x} = \tan x$ ($0 < x < \frac{\pi}{2}$), $10\sin^4\theta + 15\cos^4\theta = 6$, $\sin\theta \frac{1}{\sqrt{3}} - \cos\theta \frac{1}{2} = 0$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Trigonometric Ratio And Identities — Core Principles & Governing Laws (Very High Yield — 37 PYQs)

Core Principle: Fundamental definitions, core postulates, and governing physical/chemical relations in Trigonometric Ratio And Identities.

■ **Caution / JEE Pitfall:** Carefully verify units, standard reference states, and sign conventions in Trigonometric Ratio And Identities.

■ **Speed Trick:** Focus on high-yield formulas and standard limiting approximations tested in Trigonometric Ratio And Identities PYQs.

Representative Source PYQs:

- [2026 Morning Shift] Q#1 (MCQ): "Let $\tan A, \tan B$, where $A, B \in (-\pi/2, \pi/2)$, be the roots of the quadratic equation $x^2 - 2x - 5 = 0$. Then $20\sin^2\{ \dots \}$ " [Ans: C]
- [2026 Morning Shift] Q#1 (Numerical): "If $A = \frac{\sin 3^\circ \cos 9^\circ + \frac{\sin 9^\circ \cos 27^\circ + \frac{\sin 27^\circ \cos 81^\circ}{\dots}}{\dots}$ and $B = \tan 81^\circ - \tan 3^\circ$ " [Ans: 2]
- [2026 Evening Shift] Q#2 (MCQ): "Let $P = \theta \in [0, 4\pi]: \tan^2 \theta \neq 1$ and $S = a \in \dots: 2(\cos^8 \theta - \sin^8 \theta) \sec 2\theta = a^2, \theta \in P$. Then $n(S)$ " [Ans: N/A]

3.2 Trigonometric Ratio And Identities — Standard Calculations & Problem Archetypes (Very High Yield — 32 PYQs)

Core Principle: Quantitative numerical problem-solving techniques and standard question patterns frequently seen in Trigonometric Ratio And Identities.

■ **Caution / JEE Pitfall:** Watch for algebraic rounding, stoichiometric balancing, and boundary condition checks.

■ **Speed Trick:** Dimensional consistency checks and ratio comparisons can quickly eliminate incorrect options.

Representative Source PYQs:

- [2026 Morning Shift] Q#1 (MCQ): "The value of $\frac{\sqrt{3} \csc 20^\circ - \sec 20^\circ \cos 20^\circ \cos 40^\circ \cos 60^\circ \cos 80^\circ}{\dots}$ is equal to..." [Ans: N/A]
- [2026 Morning Shift] Q#2 (MCQ): "If $\cot x = (5/12)$ for some $x \in (\pi, 3\pi/2)$, then $\sin 7x (\cos \frac{13x}{2} + \sin \frac{13x}{2}) + \cos 7x (\cos \frac{13x}{2} - \sin \frac{13x}{2})$ " [Ans: N/A]
- [2026 Morning Shift] Q#2 (Numerical): "Let $\vec{a}(k) = (\tan \theta(k))\hat{i} + \hat{j}$ and $\vec{b}(k) = \hat{i} - (\cot \theta(k))\hat{j}$, where $\theta_k = \frac{2^{k-1}\pi}{2^n}$ " [Ans: 3]

3.3 Trigonometric Ratio And Identities — Applied Cases, Reactions & Traps (Standard Yield — 0 PYQs)

Core Principle: Special cases, experimental observations, and conceptual traps designed to test deep understanding of Trigonometric Ratio And Identities.

■ **Caution / JEE Pitfall:** Read questions carefully for keywords like 'incorrect', 'not true', or non-standard conditions.

■ **Speed Trick:** Elimination of extreme or unphysical answer choices improves accuracy in assertion-reason and multiple-statement questions.

Representative Source PYQs:

- **[2026 Morning Shift] Q#1 (MCQ):** "Let $\tan A, \tan B$, where $A, B \in (-\pi/2, \pi/2)$, be the roots of the quadratic equation $x^2 - 2x - 5 = 0$. Then $20\sin^2((A+B)/2)$ [Ans: C]
- **[2026 Morning Shift] Q#1 (MCQ):** "The value of $\frac{\sqrt{3}\operatorname{cosec}20^\circ - \sec20^\circ \cos20^\circ \cos40^\circ \cos60^\circ \cos80^\circ}{\cos20^\circ}$ is equal to..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 5th April Morning Shift (Q#1)

Let $\tan A, \tan B$, where $A, B \in (-\pi/2, \pi/2)$, be the roots of the quadratic equation $x^2 - 2x - 5 = 0$. Then $20\sin^2((A+B)/2)$ is equal to:

- (A) $10 + \sqrt{10}$
- (B) $10 - 2\sqrt{10}$
- (C) $10 - 3\sqrt{10}$ [Correct]
- (D) $10 - \sqrt{10}$

Correct Answer: C

Solution Analysis: Let the roots of the equation $x^2 - 2x - 5 = 0$ be $\tan A$ and $\tan B$. So, by using relations between roots and coefficients, $\tan A + \tan B = 2$ and $\tan A \tan B = -5$. We have to find $20\sin^2((A+B)/2)$. First, find $\tan(A+B)$ using the formula $\tan(A+B) = (\tan A + \tan B) / (1 - \tan A \tan B)$. Substituting the values: $\tan(A+B) = (2 / (1 - (-5))) = (2 / 6) = (1 / 3)$. Now let $\theta = A+B$. Then $\tan \theta = (1 / 3)$. We need $\sin^2(\theta/2)$. Using the identity $\sin^2(\theta/2) = (1 - \cos \theta) / 2$, we first find $\cos \theta$. Since $\tan \theta = (1 / 3)$, take a right triangle with

Benchmark Problem 2: JEE Main 2026 (Online) 4th April Morning Shift (Q#2)

Let $\vec{a}_k = (\tan \theta_k) \hat{i} + \hat{j}$ and $\vec{b}_k = \hat{i} - (\cot \theta_k) \hat{j}$, where $\theta_k = \frac{2^k - 1}{2^{n+1}} \pi$, for some $n \in \mathbb{N}, n > 5$. Then the value of $\frac{\sum_{k=1}^n |\vec{a}_k|^2 \sum_{k=1}^n |\vec{b}_k|^2}{\sum_{k=1}^n |\vec{a}_k| |\vec{b}_k|}$ is

Correct Answer: 3

Solution Analysis: $\vec{a}_k = (\tan \theta_k) \hat{i} + \hat{j}$, $\vec{b}_k = \hat{i} - (\cot \theta_k) \hat{j}$, where $\theta_k = \frac{2^k - 1}{2^{n+1}} \pi, k = 1, 2, \dots, n$. We have to find $\frac{\sum_{k=1}^n |\vec{a}_k|^2 \sum_{k=1}^n |\vec{b}_k|^2}{\sum_{k=1}^n |\vec{a}_k| |\vec{b}_k|}$. Now let us compute the magnitudes. For $\vec{a}_k$, $|\vec{a}_k|^2 = (\tan \theta_k)^2 + 1 = \sec^2 \theta_k$. For $\vec{b}_k$, $|\vec{b}_k|^2 = 1 + (\cot \theta_k)^2 = \csc^2 \theta_k$. So the required ratio becomes $\frac{\sum_{k=1}^n \sec^2 \theta_k \sum_{k=1}^n \csc^2 \theta_k}{\sum_{k=1}^n \sec \theta_k \csc \theta_k}$. Now use $\sec^2 \theta = \frac{1}{\cos^2 \theta}, \csc^2 \theta = \frac{1}{\sin^2 \theta}$